higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# NATIONAL CERTIFICATE (VOCATIONAL)

## ENGINEERING TECHNOLOGY
## NQF LEVEL 2

(6021012)

**1 March 2019 (X-Paper)**
**09:00–12:00**

**This question paper consists of 9 pages and 1 addendum.**

NCV

**TIME: 3 HOURS**
**MARKS: 100**

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**INSTRUCTIONS AND INFORMATION**

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL sketches must be neat, clean and in proportion.
5. Clearly cross out ALL work you do NOT want to be marked.
6. Start each question on a NEW page.
7. Write neatly and legibly.

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## QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 An external micrometer is used to measure …

A inside measurements of a component.
B outside measurements only.
C perpendicular lines.
D the quantity of air delivered by a compressor.

1.1.2 A useful instrument with three heads that are used separately and held to a beam by a nut:

A Surface plate
B Jenny calliper
C Combination set
D Dividers

1.1.3 A hydrometer is used to measure the …

A level of liquid in a glass.
B density of a liquid.
C pressure in a liquid.
D acid in a battery.

1.1.4 Which ONE of the following lines can be used to indicate the hidden detail in first-angle orthographic projection:

A Object lines
B Dimension lines
C Centre lines
D Broken lines

1.1.5 Identify the correct parts of an angle grinder:

A Table and base
B Chuck and trigger
C Guard and disc
D Hammer and nail

(5 × 1) (5)

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1.2 Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–F) next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

| COLUMN A | | COLUMN B | |
|---|---|---|---|
| 1.2.1 | Pop riveter | A | protects the user of a machine |
| 1.2.2 | Tin snips | B | designed to fasten two parts together |
| 1.2.3 | Pin punch | C | used to cut thin sheet metal |
| 1.2.4 | Wheel guard | D | used to start driving out a pin |
| 1.2.5 | Ram | E | guided shaft used to exert pressure |
| | | F | hollow shaft that fits over another shaft |

(5 × 1) (5)

1.3 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

1.3.1 A V-block is used to support square bars.

1.3.2 A datum line is a fixed starting point.

1.3.3 Measuring is how one inspects with the eye.

1.3.4 Tolerance is not a variation of a nominal dimension.

1.3.5 Slip gauges are highly accurate blocks stacked together to give the required length to be measured.

(5 × 1) (5)

1.4 Give ONE term for each of the following descriptions. Write only the term next to the question number (1.4.1–1.4.5) in the ANSWER BOOK.

1.4.1 Action of a power tool when it jumps backwards as a result of the blade or drill jamming in the workpiece

1.4.2 Clamping device on a drilling machine which securely holds a drill bit when drilling holes

1.4.3 Rate at which work is done over a period of time

1.4.4 Wheel with a groove rim used to transmit power by means of a belt or rope

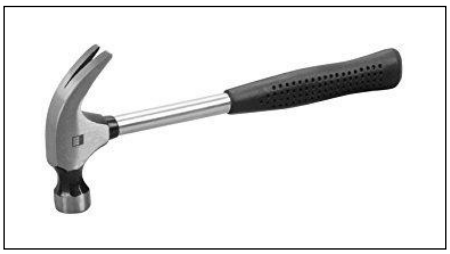
1.4.5 Reduction in the volume of air due to pressure exerted on it

(5 × 1) (5)

**[20]**

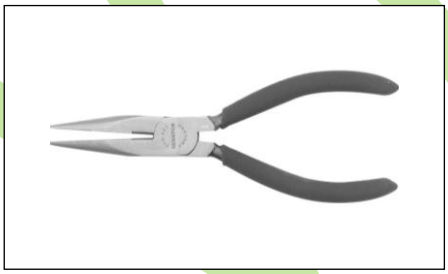
## QUESTION 2

2.1 Name each of the following hand tools. Write only the answer next to the letter (A–F) in the ANSWER BOOK.

A 

D 

B 

E 

C 

F 

(6 × 1) (6)

2.2 Give FOUR reasons for hacksaw-blade failures. (4)

2.3 Name the tool one would use to check the gap on a spark plug. (1)

2.4 Make a neat labelled sketch of a hacksaw. Insert FOUR labels on the sketch. (6)

**[17]**

## QUESTION 3

3.1 Refer to FIGURE 1 and answer the questions.

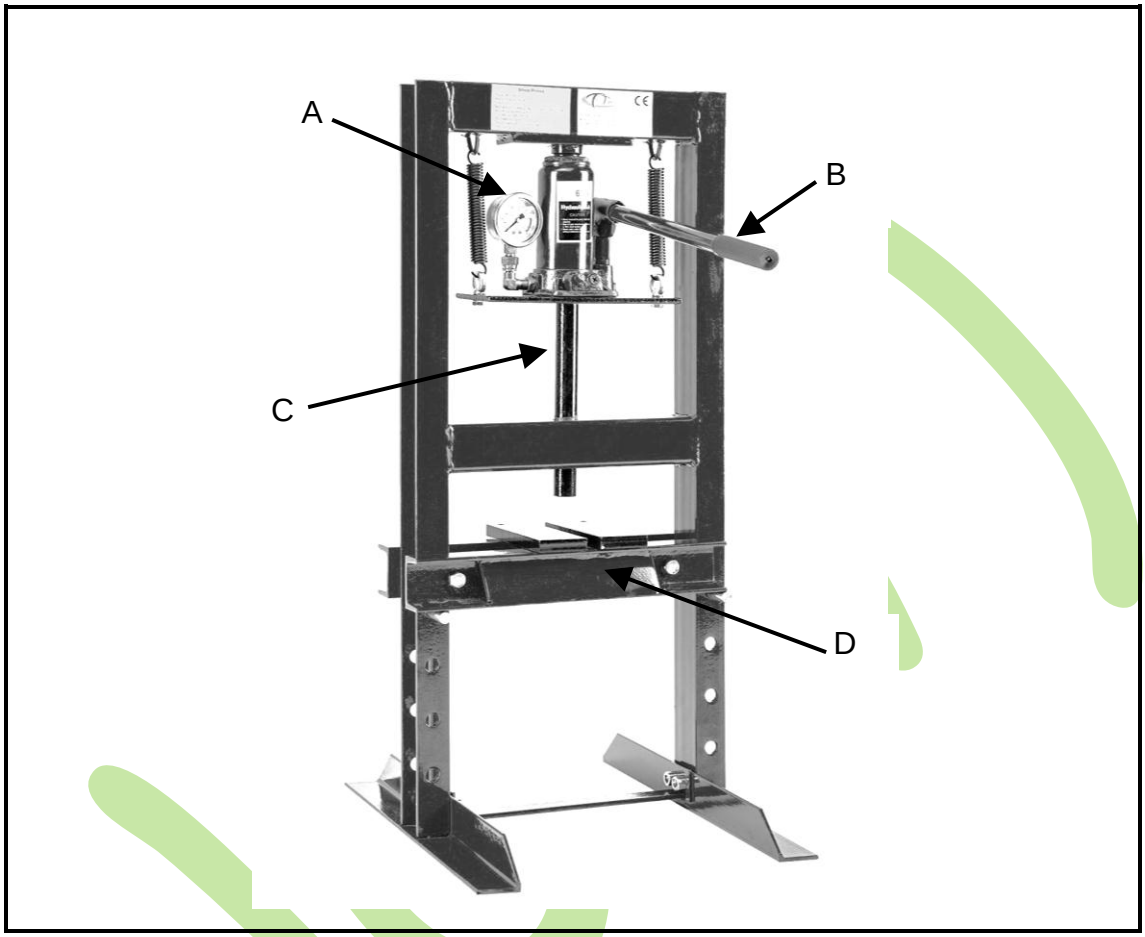



**FIGURE 1**

3.1.1 Name the power tool shown in FIGURE 1. (1)

3.1.2 Name the parts of the power tool by writing only the answer next to the letter (A–D) in the ANSWER BOOK. (4 × 1) (4)

3.2 List FIVE important factors that determine the nature of a grinding wheel. (5)

3.3 State TWO safety measures when using each of the following power tools:

3.3.1 Portable drilling machine

3.3.2 Pedestal drilling machine

(2 × 2) (4)

**[14]**

## QUESTION 4

4.1 Name THREE groups of gauges. (3)

4.2 State the purpose of each gauge mentioned in QUESTION 3.1. (3)

4.3 State TWO uses of each of the following measuring tools:

4.3.1 Calliper (inside or outside)

4.3.2 Straight edge

(2 × 2) (4)

4.4 State THREE factors to consider when caring for and maintaining a thread-pitch gauge. (3)

**[13]**

## QUESTION 5

5.1 FIGURE 2 shows a precision-measuring tool.

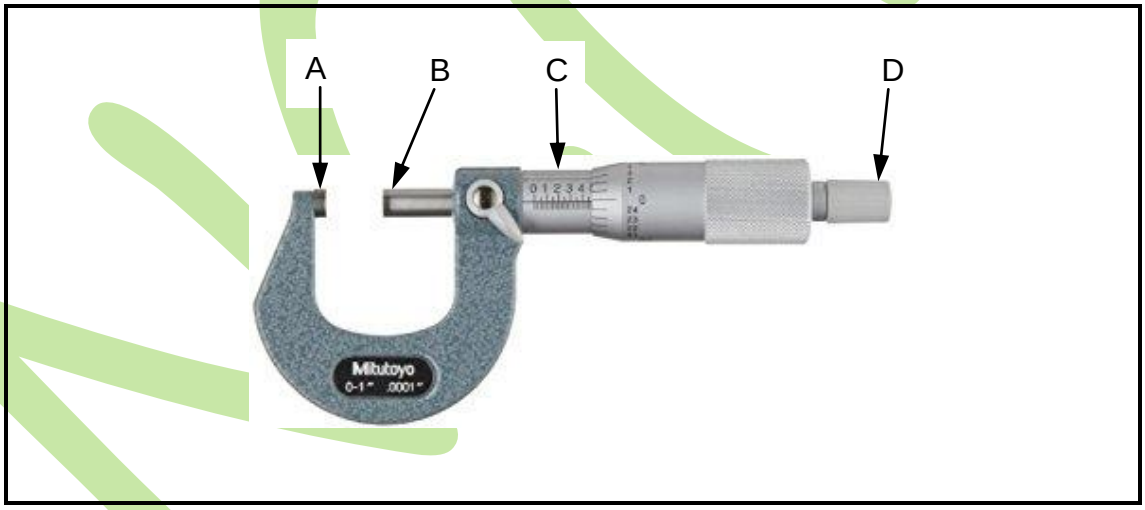



**FIGURE 2**

5.1.1 Name the precision-measuring tool shown in FIGURE 2. (1)

5.1.2 Label the parts of the precision-measuring tool by writing only the answer next to the letter (A–D) in the ANSWER BOOK. (4 × 1) (4)

5.2 Explain in THREE steps how to test the accuracy of a 0-25 mm outside micrometre without using a test bar. (3)

**[8]**

## QUESTION 6

6.1 An object has a mass of 68 kg.

Calculate the weight of the object.

**HINT:** W = m × g (3)

6.2 Write the symbols for each of the following units:

6.2.1 Kelvin

6.2.2 Ampere

6.2.3 Mole

(3 × 1) (3)

6.3 Convert each of the following:

6.3.1 900 g = … kg

6.3.2 3 500 mℓ = … ℓ

(2 × 1) (2)

6.4 FIGURE 3 shows a marking-off tool.

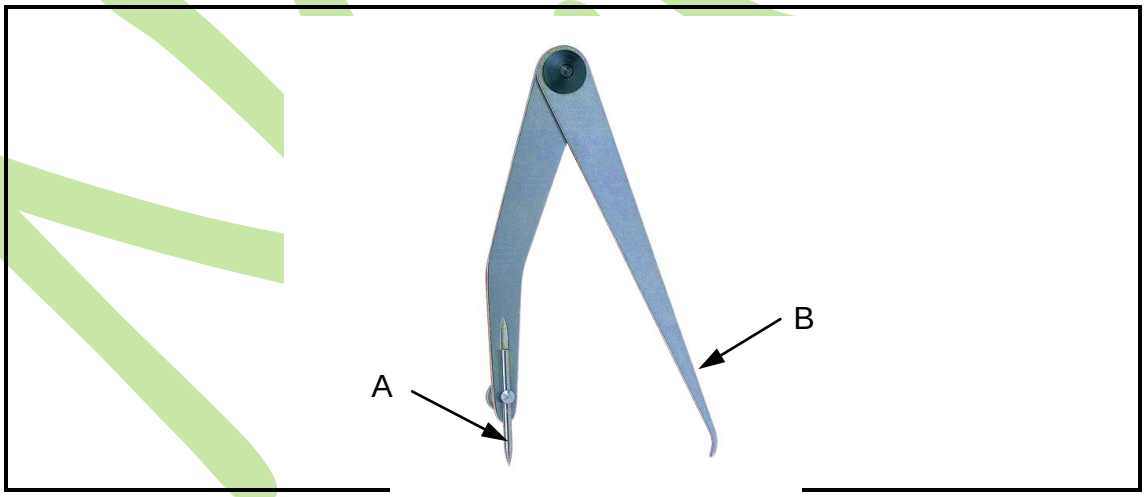



**FIGURE 3**

6.4.1 Name the marking-off tool shown in FIGURE 3. (1)

6.4.2 Name the parts of the marking-off tool by writing only the answer next to the letter (A–B) in the ANSWER BOOK. (2 × 1) (2)

6.4.3 State TWO functions of the marking-off tool shown in FIGURE 3. (2)

**[13]**

**QUESTION 7**

FIGURE 4 on the ADDENDUM (attached) shows an isometric view of a machine part.

Draw, to scale 1:1, in first-angle orthographic projection each of the following:

| | | |
|---|---|---|
| 7.1 | Front view (A) | (4) |
| 7.2 | Top view (B) | (3) |
| 7.3 | Left view (C) | (3) |
| 7.4 | Insert ONE correct dimension per view. | (3) |
| 7.5 | Insert the projection symbol in the space below the left view. | (2) |
| | | **[15]** |

**TOTAL: 100**

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**ADDENDUM**

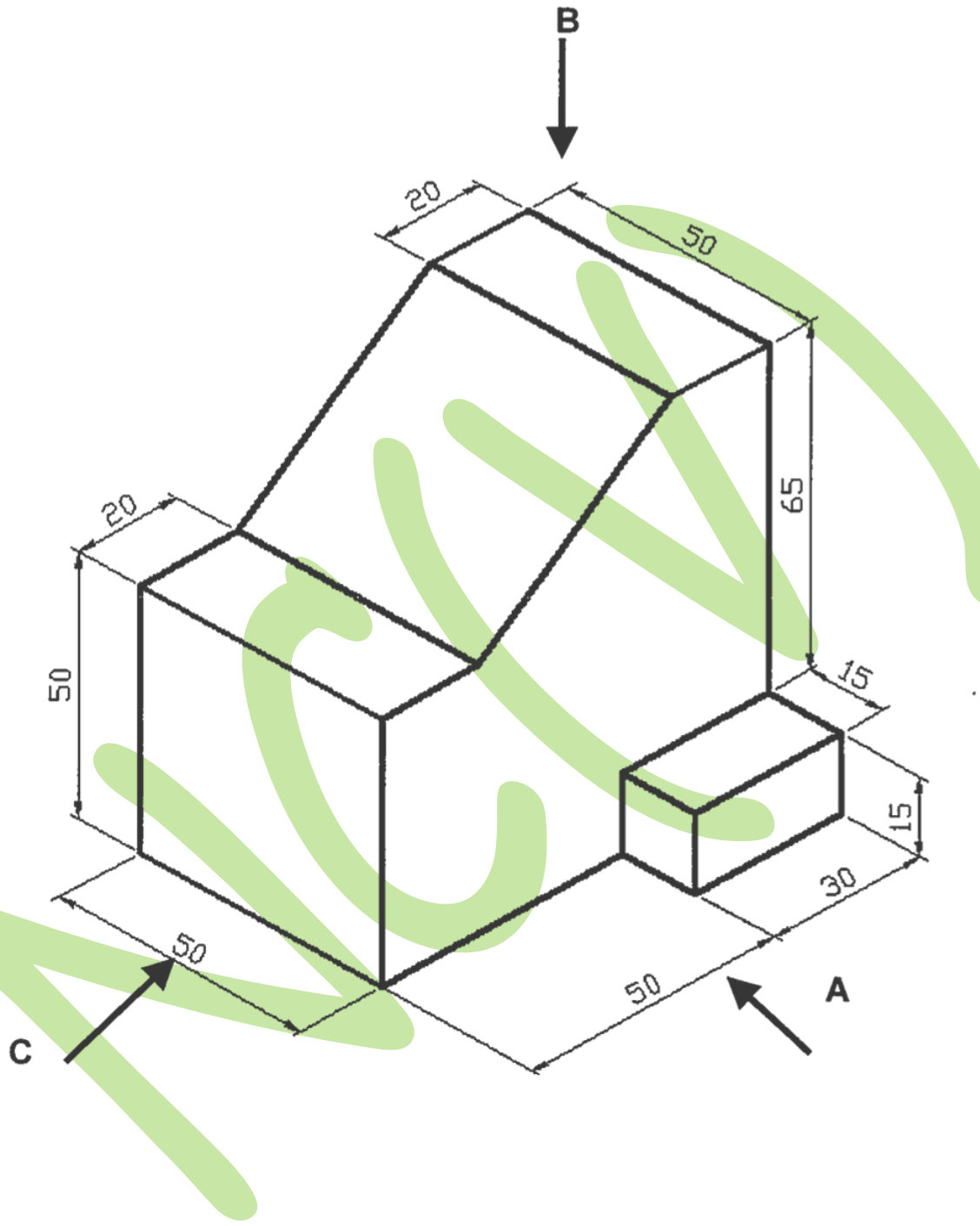



**FIGURE 4**

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